NSCAS NEBRASKA STUDENT-CENTERED ASSESSMENT SYSTEM

Grade 8
General
Mathematics

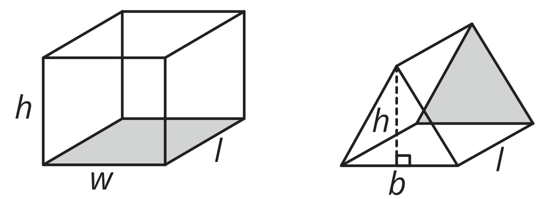
NSCAS NEBRASKA STUDENT-CENTERED ASSESSMENT SYSTEM

Mathematics Reference Sheet

2-Dimensional Shape	Area	Circumference
Circle	$A = \pi r^2$	$C = \pi d = 2\pi r$
Triangle	$A = \frac{1}{2}bh$	**Perimeter**
Rectangle	$A = l \times w$	$P = 2l + 2w$ $= 2(l + w)$
Square	$A = s \times s$	$P = s + s + s + s$
Trapezoid	$A = \frac{1}{2}h(b_1 + b_2)$	
Parallelogram	$A = bh$	

3-Dimensional Shape	Volume	Surface Area
Rectangular Prism	$V = lwh = Bh$	SA = the sum of the areas of all the faces
Triangular Prism	$V = \frac{1}{2}bhl = Bl$	SA = the sum of the areas of all the faces
Cone	$V = \frac{1}{3}\pi r^2 h$	
Cylinder	$V = \pi r^2 h$	
Sphere	$V = \frac{4}{3}\pi r^3$	

In the images to the right, the shaded faces are the bases.



2-Dimensional Shape Key		
b = base length	l = length	s = side
h = height	w = width	
d = diameter	r = radius	

Use 3.14 for π.

3-Dimensional Shape Key		
d = diameter	l = length	w = width
r = radius	h = height of shape	B = area of base
	b = base length	

Percent Change	Pythagorean Theorem
% change = $\frac{\text{difference in amounts}}{\text{original amount}}$	$c^2 = a^2 + b^2$

Standard Units	Metric Units
Conversions – Length	
1 foot (ft) = 12 inches (in.)	1 centimeter (cm) = 10 millimeters (mm)
1 yard (yd) = 3 feet (ft) = 36 inches (in.)	1 meter (m) = 100 centimeters (cm)
1 mile (mi) = 1,760 yards (yd) = 5,280 feet (ft)	1 meter (m) = 1,000 millimeters (mm)
	1 kilometer (km) = 1,000 meters (m)
Conversions – Volume	
1 cup = 8 fluid ounces (fl oz)	1 liter (l) = 1,000 milliliters (ml)
1 pint (pt) = 2 cups	1 liter (l) = 1,000 cubic centimeters (cu cm)
1 quart (qt) = 2 pints (pt)	
1 gallon (gal) = 4 quarts (qt)	
Conversions – Weight/Mass	
1 pound (lb) = 16 ounces (oz)	1 gram (g) = 1,000 milligrams (mg)
1 ton = 2,000 pounds (lb)	1 kilogram (kg) = 1,000 grams (g)

Directions:

On the following pages of your practice test booklet are questions for the Grade 8 *Nebraska Student-Centered Assessment System (NSCAS) General Mathematics* practice test.

Read these directions carefully before beginning the test.

This practice test will include several different types of questions. Multiple choice questions will ask you to select an answer from among four choices. Multiple select questions will ask you to select multiple correct answers from among five or more answer choices. For some questions, there may be two parts, Part A and Part B, where each part has a multiple choice or multiple select question. Some questions will ask you to construct an answer by following the directions given.

For all questions:

- Read each question carefully and choose the best answer.
- You may use scratch paper to solve the problems.
- The Mathematics Reference Sheet is provided in the back of the test booklet. You may refer to this page at any time during the test.
- Be sure to answer ALL the questions.

STOP.

1. Aaron sells jackets online. He adds \$4.75 per jacket onto every purchase for shipping. He collected a total of \$188.75 selling 5 jackets. Which equation could be used to calculate the sale price, j, of each jacket?

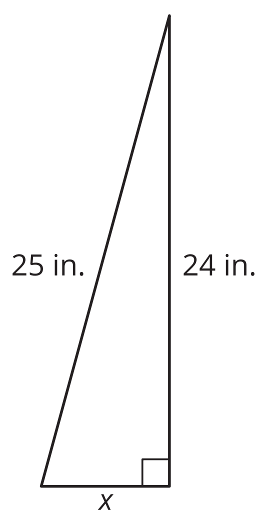
   A. $5(j + 4.75) = 188.75$

   B. $5j + 4.75 = 188.75$

   C. $5(4.75) + j = 188.75$

   D. $(5 + 4.75)j = 188.75$

2. **Use the figure below to answer the question.**



   Using the Pythagorean theorem, what is the value of x?

   A. 1 inch

   B. 7 inches

   C. 41 inches

   D. 49 inches

3. **This question has two parts. Answer Part A, and then answer Part B.**

Part A

Which equation has exactly one solution?

A. $3x + 4 = 5x + 4$

B. $6x + 2 = 6x + 3$

C. $7x + 1 = 2x + 5x + 1$

D. $10x + 6 = 2(5x + 3)$

Part B

Which statements explain why the equation has exactly one solution?

Select **all** that apply.

A. The equation simplifies to eliminate the *x*'s and have different constants on both sides of the equation.

B. The equation simplifies to eliminate the *x*'s and have the same constant on both sides of the equation.

C. The equation simplifies to *x*'s on one side of the equation and a constant on the other side.

D. Substituting the number one into the equation makes the equation true.

E. There is only one value for *x* that makes the equation true.

F. There is only one variable in the equation.

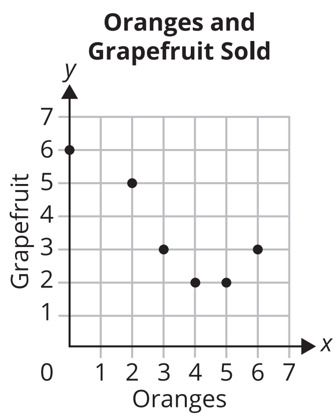
Go on to the next page.

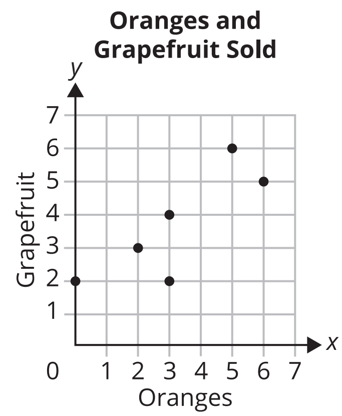
4. **Use the table below to answer the question.**

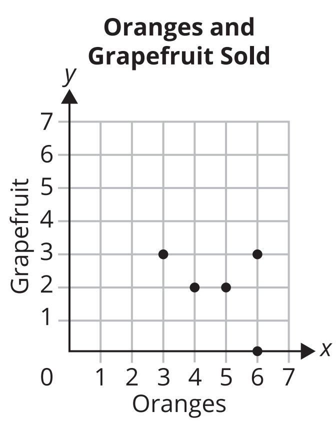
Oranges and Grapefruit Sold

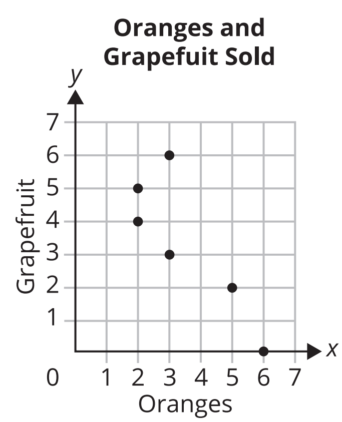
Oranges	3	4	0	2	5	6
Grapefruit	3	2	6	5	2	3

The table shows the numbers of oranges and grapefruits sold to six different customers. Which scatter plot represents the data?

A. 

B. 

C. 

D. 

5. Which list is ordered from LEAST to GREATEST?

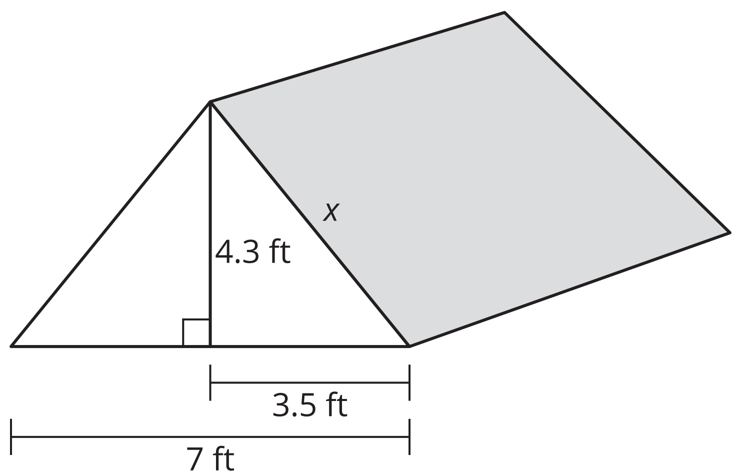
A. $-7.6,\ -\frac{76}{12},\ \sqrt{18},\ 3.99$

B. $-\frac{76}{12},\ -7.6,\ 3.99,\ \sqrt{18}$

C. $-7.6,\ -\frac{76}{12},\ 3.99,\ \sqrt{18}$

D. $-\frac{76}{12},\ -7.6,\ \sqrt{18},\ 3.99$

6. **Use the figure to answer the question.**



A company wants to make a tent that has a height of 4.3 feet and a total width of 7 feet. The company needs to know the length of edge *x* to know how much fabric is needed for the side of the tent. To the nearest tenth, what is the length of edge *x* of the tent? Write the answer in the box.

[ ] feet

7. Which value of y makes $\frac{4(y - 3)}{2} = 9 - y$ true?

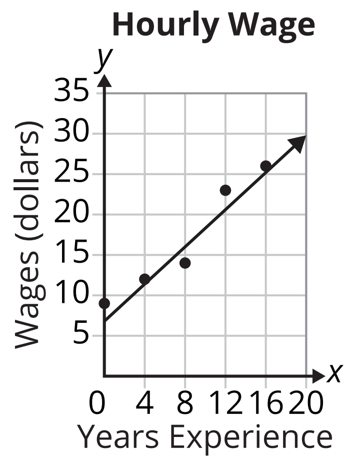
A. 3

B. 4

C. 5

D. 6

8. The line shows what people in a particular job can expect to earn based on their years of experience.



Which statements about the information in the graph are true?

Select **all** that apply.

A. An employee with 4 years of experience can expect to earn approximately \$12 per hour.

B. An employee earning \$20 per hour has approximately 11 years of experience.

C. With 16 years of experience, an employee can expect to earn \$30 per hour.

D. With 20 years of experience, an employee can expect to earn \$35 per hour.

E. An employee can expect a raise of slightly more than \$1 per hour each year.

F. An employee can expect to earn \$10 per hour as a starting wage.

9. What is the value of $|3 + 5| - |4|$?

   A. -12

   B. -4

   C. 4

   D. 12

10. A ball has a diameter of 12 inches. What is the volume of the ball? Use 3.14 for π.

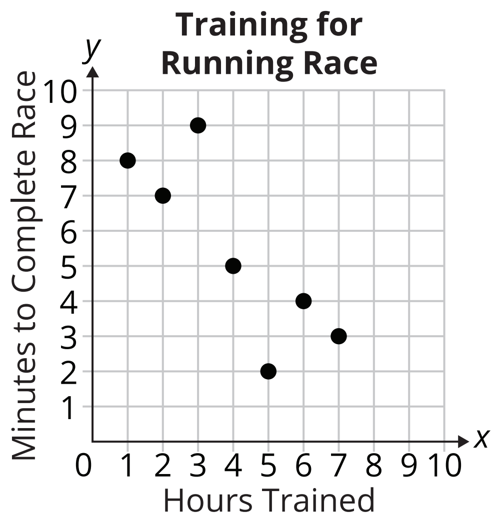
   A. 150.72 cubic inches

   B. 904.32 cubic inches

   C. 2,712.96 cubic inches

   D. 7,234.56 cubic inches

11. **Use the scatter plot to answer the question.**



Which set of data was used to create the scatter plot? Select and write the numbers to complete the missing values in the table.

1 2 3 4 5 6 7 8 9 10

Training for Running Race

Hours Trained	3		1	4	5		6
Minutes to Complete Race	9	7	8		2	3	

Go on to the next page.

12. **Use the table to answer the question.**

x	y
-2	12
0	7
2	2

What is the constant rate of change in the table?

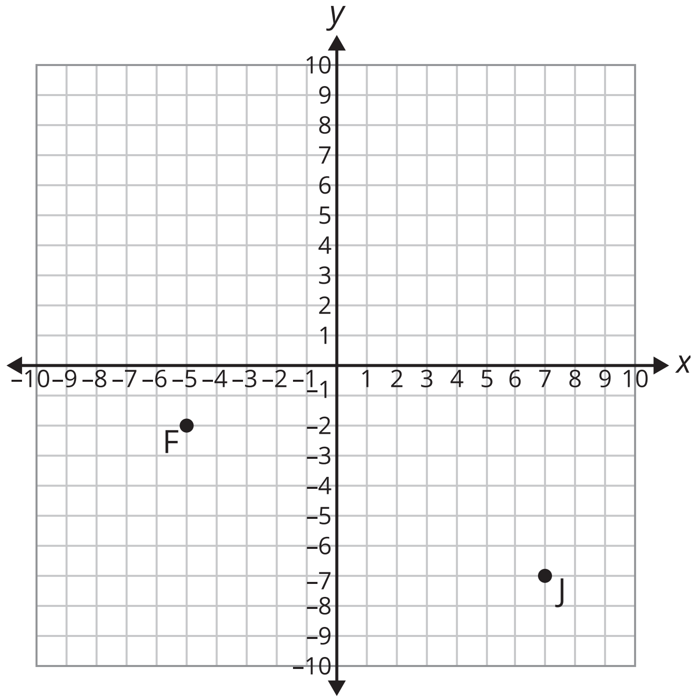
A. $-\frac{5}{2}$

B. $-\frac{2}{5}$

C. $\frac{2}{5}$

D. $\frac{5}{2}$

13. **Use the points on the coordinate plane below to answer the question.**



What is the distance between points F and J?

A. 5

B. 13

C. 17

D. 30

14. **This question has two parts. Answer Part A, and then answer Part B.**

Part A

What is -0.0000012 written in scientific notation?

A. -1.2×10^{-6}

B. -1.2×10^{-5}

C. -1.2×10^{5}

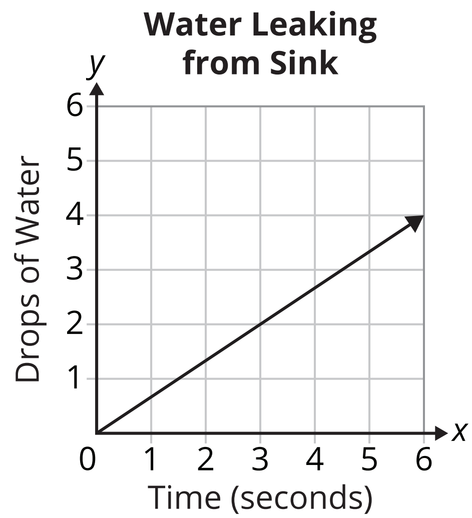
D. -1.2×10^{6}

Part B

What is 8.9×10^{4} written in standard notation? Write the answer in the box.

Go on to the next page.

15. **Use the graph below to answer the question.**



The graph shows the amount of water leaking from a sink over time. Which statement correctly compares the amount of water the sink leaks and the time?

A. For every 1 second, the sink leaks 2 drops of water.

B. For every 2 seconds, the sink leaks 1 drop of water.

C. For every 2 seconds, the sink leaks 3 drops of water.

D. For every 3 seconds, the sink leaks 2 drops of water.

16. What is the square root of 100?

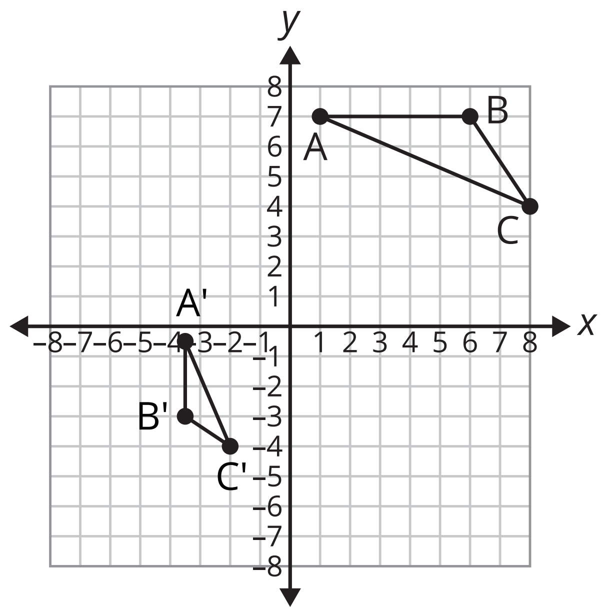
A. 5

B. 10

C. 25

D. 50

17. **Use the graph to complete the statements.**



Triangle ABC is dilated, rotated, and reflected, in that order, to map onto triangle A' B' C'. Which dilation, rotation, and reflection transform triangle ABC onto triangle A' B' C'? Select the correct choices to complete the statements.

The first transformation is a dilation centered at the origin with a scale factor of ($\mathbf{\frac{1}{3}}$ $\mathbf{\frac{1}{2}}$ **2** **3**).

The second transformation is a rotation (**90°** **180°** **360°**) clockwise about the origin.

The third transformation is a reflection over the (***x*-axis**, ***y*-axis**, **line x = 1**, **line *y* = 4**).

18. What is the value of x in the equation $\frac{7}{3}x = \frac{1}{3}x - 24$? Write the answer in the box.

$x =$ ☐

19. What is the value of $(-4)^2$?

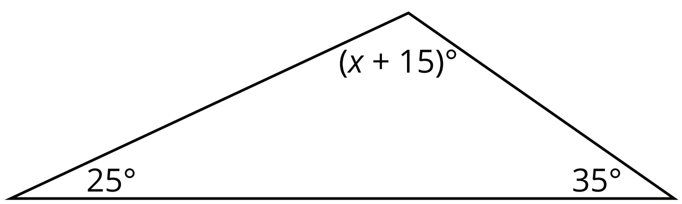
A. -16

B. $-\frac{1}{16}$

C. $\frac{1}{16}$

D. 16

20. **Use the diagram below to answer the question.**



What is the value of *x* in the triangle?

A. 60

B. 75

C. 105

D. 120

NSCAS General Grade 8
Practice Test Answer Key
Mathematics (English)

<table>
<tr><th>Sequence</th><th>Key</th><th>Points</th></tr>
<tr><td>1</td><td>A</td><td>1</td></tr>
<tr><td>2</td><td>B</td><td>1</td></tr>
<tr><td rowspan="2">3</td><td>Part A: A
Part B: C, E</td><td>2</td></tr>
<tr><td>Part A or Part B</td><td>1</td></tr>
<tr><td>4</td><td>A</td><td>1</td></tr>
<tr><td>5</td><td>C</td><td>1</td></tr>
<tr><td>6</td><td>5.5 or equivalent</td><td>1</td></tr>
<tr><td>7</td><td>C</td><td>1</td></tr>
<tr><td rowspan="2">8</td><td>A, B, E</td><td>2</td></tr>
<tr><td>2 of 3 correct responses</td><td>1</td></tr>
<tr><td>9</td><td>C</td><td>1</td></tr>
<tr><td>10</td><td>B</td><td>1</td></tr>
<tr><td>11</td><td><table><tr><td>Hours Trained</td><td>3</td><td>2</td><td>1</td><td>4</td><td>5</td><td>7</td><td>6</td></tr><tr><td>Minutes to Complete Race</td><td>9</td><td>7</td><td>8</td><td>5</td><td>2</td><td>3</td><td>4</td></tr></table></td><td>1</td></tr>
<tr><td>12</td><td>A</td><td>1</td></tr>
<tr><td>13</td><td>B</td><td>1</td></tr>
<tr><td rowspan="2">14</td><td>Part A: A
Part B: 89000 or equivalent</td><td>2</td></tr>
<tr><td>Part A or Part B</td><td>1</td></tr>
<tr><td>15</td><td>D</td><td>1</td></tr>
<tr><td>16</td><td>B</td><td>1</td></tr>
<tr><td rowspan="2">17</td><td>$\frac{1}{2}$
90°
y-axis</td><td>2</td></tr>
<tr><td>2 of 3 correct responses</td><td>1</td></tr>
<tr><td>18</td><td>–12 or equivalent</td><td>1</td></tr>
<tr><td>19</td><td>D</td><td>1</td></tr>
<tr><td>20</td><td>C</td><td>1</td></tr>
</table>